1. NAME OF THE MEDICINAL PRODUCT

Vaxigrip suspension for injection in pre-filled syringe

Trivalent influenza vaccine (split virion, inactivated)

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Influenza virus (inactivated, split) of the following strains*:

- A/Missouri/11/2025 (H1N1)pdm09-like strain (A/Switzerland/6849/2025, IVR-278) .... 15 micrograms HA**
- A/Singapore/GP20238/2024 (H3N2)-like strain (A/Singapore/GP20238/2024, IVR-277) . ... 15 micrograms HA**
- B/Austria/1359417/2021-like strain (B/Michigan/01/2021, wild type)................... ……… .... 15 micrograms HA**

Per 0.5 mL dose

*	Propagated in fertilised hens' eggs from healthy chicken flocks

**	Haemagglutinin

This vaccine complies with the WHO recommendations (Southern Hemisphere) for the 2026 season.

For the full list of excipients, see Section 6.1.

Vaxigrip may contain traces of eggs, such as ovalbumin, and of neomycin, formaldehyde and octoxynol-9, which are used during the manufacturing process (see Section 4.3).

3. PHARMACEUTICAL FORM

Suspension for injection.

The vaccine, after shaking gently, is a colourless opalescent liquid.

4. CLINICAL PARTICULARS

4.1 Therapeutic Indications

Vaxigrip is indicated for the prevention of influenza disease caused by the two influenza A virus subtypes and the influenza B virus type contained in the vaccine for:

- active immunisation of adults, including pregnant women, and children from 6 months of age and older,
- passive protection of infant(s) from birth to less than 6 months of age following vaccination of pregnant women (see Sections 4.4, 4.6 and 5.1).

The use of Vaxigrip should be based on official recommendations on vaccination against influenza.

4.2 Posology and Method of Administration

<u>Posology</u>

Adults: one dose of 0.5 mL

Paediatric population

- Children from 6 months to 17 years of age: one dose of 0.5 mL.

For children less than 9 years of age who have not previously been vaccinated, a second dose of 0.5 mL should be given after an interval of at least 4 weeks.

- Infants less than 6 months of age: the safety and efficacy of Vaxigrip administration (active immunisation) have not been established. No data are available.

  Regarding passive protection: one 0.5 mL dose given to pregnant women may protect infants from birth to less than 6 months of age (see Sections 4.4, 4,6 and 5.1).

Method of Administration

The preferred route of administration for this vaccine is intramuscular although it can also be given subcutaneously.

The preferred sites for intramuscular injection are the anterolateral aspect of the thigh (or the deltoid muscle if muscle mass is adequate) in children 6 months through 35 months of age, or the deltoid muscle in children from 36 months of age and adults.

Precautions to be taken before handling or administering the medicinal product.

For instructions on preparation of the medicinal product before administration, see Section 6.6.

4.3 Contraindications

Hypersensitivity to the active substances, to any of the excipients listed in Section 6.1 or to any component that may be present as traces such as eggs (ovalbumin, chicken proteins), neomycin, formaldehyde and octoxynol-9.

4.4 Special Warnings and Precautions for Use

Traceability
In order to improve the traceability of biological medicinal products, the name and the batch number of the administered product should be clearly recorded.

Hypersensitivity
As with all injectable vaccines, appropriate medical treatment and supervision should always be readily available in case of an anaphylactic reaction following the administration of the vaccine.

Concurrent illness
Vaccination should be postponed in patients with acute febrile illness until the fever is resolved.

Precautions for use
Vaxigrip should under no circumstances be administered intravascularly.

Thrombocytopenia and coagulation disorders
As with other vaccines administered intramuscularly, the vaccine should be administered with caution to subjects with thrombocytopenia or a coagulation disorder since bleeding may occur following an intramuscular administration to these subjects.

Syncope

Syncope (fainting) can occur following, or even before, any vaccination as a psychogenic response to the needle injection. Procedures should be in place to prevent injury from fainting and manage syncopal reactions.

Protection
Vaxigrip is intended to provide protection against those strains of influenza virus from which the vaccine is prepared.
As with any vaccine, vaccination with Vaxigrip may not protect all vaccinees.
Regarding passive protection, not all infants less than 6 months of age born to women vaccinated during pregnancy will be protected (see Section 5.1).

Immunodeficiency
Antibody response in patients with congenital or acquired immunodeficiency may be insufficient.

Potassium and sodium content
This medicinal product contains less than 1 mmol (39 mg) of potassium and less than 1 mmol (23 mg) of sodium per dose, i.e. it is essentially "potassium-free" and "sodium-free".

4.5 Interactions with Other Medicinal Products and Other Forms of Interaction

Vaxigrip could be given at the same time as other vaccines if needed.

Data showing that Vaxigrip can be administered concomitantly with other vaccines are available for the following vaccines: pneumococcal polysaccharide vaccine, tetanus, diphtheria, pertussis and polio vaccine (DTaP/IPV, Repevax), and shingles vaccine. If Vaxigrip is given at the same time as other vaccines, separate injection sites and separate syringes should be used.

The immunological response may be reduced if the patient is undergoing immunosuppressant treatment.

4.6 Fertility, Pregnancy and Lactation

Pregnancy

Pregnant women are at high risk of influenza complications, including premature labour and delivery, hospitalisation, and death: pregnant women should receive an influenza vaccine.

Vaxigrip can be used in all stages of pregnancy.

More data are available on the safety of inactivated influenza vaccines for the second and third trimesters than for the first trimester. Data from worldwide use of inactivated influenza vaccines, including Vaxigrip and Vaxigrip Tetra (quadrivalent inactivated influenza vaccine), do not indicate any adverse foetal and maternal outcomes attributable to the vaccine.

This is consistent with results observed in one clinical study in which Vaxigrip and Vaxigrip Tetra were administered in pregnant women during the second or third trimester (116 exposed pregnancies and 119 live births for Vaxigrip, 230 exposed pregnancies and 231 live births for Vaxigrip Tetra).

Data from four clinical studies with Vaxigrip administered in pregnant women during the second or third trimester (more than 5,000 exposed pregnancies and more than 5,000 live births followed up to approximately 6 months post-partum) did not indicate any adverse foetal, newborn, infant and maternal outcomes attributable to the vaccine.

In clinical studies conducted in South Africa and Nepal, there were no significant differences between the Vaxigrip and placebo groups with regards to foetal, newborn, infant and maternal outcomes (including miscarriage, stillbirth, premature birth and low birth weight).

In a study conducted in Mali, there were no significant differences between the Vaxigrip and control vaccine (quadrivalent meningococcal conjugate vaccine) groups with regards to premature births, rate of stillbirths and rates of low birth weight/small for gestational age.

For additional information, see Sections 4.8 and 5.1.

The results of one reproductive and developmental toxicity study in rabbits conducted with Vaxigrip Tetra (60 µg of total HA/dose) can be extrapolated to Vaxigrip (45 µg of total HA/dose): this study did not indicate direct or indirect harmful effects with respect to pregnancy, embryo-foetal development or early post-natal development.

Breast-feeding

Vaxigrip can be used during breast-feeding.

Fertility

No fertility data are available in humans. One animal study with Vaxigrip Tetra did not indicate harmful effects on female fertility.

4.7 Effects on Ability to Drive and Use Machines

Vaxigrip has no or negligible influence on the ability to drive and use machines.

4.8 Undesirable Effects

Summary of the safety profile

The safety profile of Vaxigrip is based on data from 46 clinical studies in which approximately 17,900 participants from 6 months of age received Vaxigrip or Vaxigrip Tetra, and data from post-marketing surveillance.

Most of the adverse reactions usually occurred within the first 3 days after vaccination and resolved spontaneously within 1 to 3 days of their onset. The intensity of most of these effects was mild to moderate.

The most frequently reported undesirable effect after vaccination in all populations including the whole group of children from 6 to 35 months of age, was injection site pain.

Tabulated list of adverse reactions

Adverse events are ranked under headings of frequency using the following convention:

Very common (≥ 1/10);

Common (≥ 1/100 to < 1/10);

Uncommon (≥ 1/1,000 to < 1/100);

Rare (≥ 1/10,000 to < 1/1,000);

Not known (cannot be estimated from available data).

Adults and the elderly

The safety profile is based on data:

- from clinical studies in more than 8,000 adults (5,064 for Vaxigrip, 3,040 for Vaxigrip Tetra) and more than 5,800 elderly subjects over 60 years of age (4,468 for Vaxigrip, 1,392 for Vaxigrip Tetra),
- from worldwide post-marketing surveillance in the general population.

In adults, the most frequently reported adverse reactions after vaccination were injection site pain (52.8%), headache (27.8%), myalgia (23.0%) and malaise (19.2%).

In the elderly, the most frequently reported adverse reactions after vaccination were injection site pain (25.8%), headache (15.6%) and myalgia (13.9%).

Table 1: Adverse reactions reported in adults and the elderly

System Organ Class (SOC)/Adverse Reactions	**Frequency**
Blood and Lymphatic System Disorders	
Lymphadenopathy (1)	Uncommon
Transient thrombocytopenia	Not known*
Immune System Disorders	
Allergic reactions such as hypersensitivity (2), atopic dermatitis (2), urticaria (2, 3), oropharyngeal pain, asthma (1), allergic rhinitis (2), rhinorrhoea (1), allergic conjunctivitis (2), pruritus (2), hot flushes (5)	Uncommon
Allergic reactions such as angioedema (2, 3), swelling of the face, erythema, rash, flushing (5), oral mucosal eruption (5), oral paraesthesia (5), throat irritation, dyspnoea (2, 3), sneezing, nasal obstruction (2), upper respiratory tract congestion (2), ocular hyperaemia (2), allergic dermatitis (2), generalised pruritus(2)	Rare
Allergic reactions such as erythematous rash, anaphylactic reaction, shock	Not known*
Metabolism and Nutrition Disorders	
Decreased appetite	Rare
Nervous System Disorders	
Headache	Very common
Dizziness (4), somnolence (4)	Uncommon

Hypoaesthesia [(2)], paraesthesia	Rare
Neuralgia, convulsions, encephalomyelitis, neuritis, Guillain Barré Syndrome	Not known*
Vascular Disorders	
Vasculitis such as Henoch-Schonlein purpura, with transient renal involvement in certain cases	Not known*
Gastrointestinal Disorders	
Diarrhoea, nausea	Uncommon
Abdominal pain [(2)], vomiting	Rare
Skin and Subcutaneous Tissue Disorders	
Hyperhidrosis [(1)]	Uncommon
Musculoskeletal and Connective Tissue Disorders	
Myalgia	Very common
Arthralgia [(1)]	Uncommon
General Disorders and Administration Site Conditions	
Injection site pain, malaise [(6)]	Very common
Fever [(7)], chills, injection site erythema, injection site induration, injection site swelling	Common
Asthenia [(1)], fatigue, injection site ecchymosis, injection site pruritus, injection site warmth, injection site discomfort	Uncommon
Flu-like symptoms, injection site exfoliation [(5)], injection site hypersensitivity [(2)]	Rare

[(1)] Rare in the elderly
[(2)] Reported during clinical trials in adults
[(3)] Not known in the elderly
[(4)] Rare in adults
[(5)] Reported during clinical trials in the elderly
[(6)] Common in the elderly
[(7)] Uncommon in the elderly

(*) Adverse reactions reported post-marketing after use of Vaxigrip or Vaxigrip Tetra

Paediatric population

The safety profile is based on data:

- from clinical studies in 1 247 children from 3 to 8 years of age (363 for Vaxigrip, 884 for Vaxigrip Tetra) and in 725 children/adolescents from 9 to 17 years of age (296 for Vaxigrip, 429 for Vaxigrip Tetra),
- from one clinical study in 1 981 children from 6 to 35 months of age (367 for Vaxigrip, 1 614 for Vaxigrip Tetra),
- from worldwide post-marketing surveillance in the general population.

Depending on immunisation history, children from 6 months to 8 years of age received one or two doses of Vaxigrip or Vaxigrip Tetra. Children/adolescents from 9 to 17 years of age received one dose.

In children from 6 months to 8 years of age, the safety profile was similar after the first and the second injections with a trend for a lower incidence of adverse reactions after the second injection compared to the first one in children from 6 to 35 months.

In children/adolescents from 9 to 17 years of age, the most frequently reported adverse reactions after vaccination were injection site pain (65.3%), myalgia (29.1%), headache (28.6%), malaise (20.3%), chills (13.0%), injection site erythema (11.7%) and injection site swelling (11.4%).

In children from 3 to 8 years of age, the most frequently reported adverse reactions after vaccination were injection site pain (59.1%), malaise (30.7%), injection site erythema (30.3%), myalgia (28.5%), headache (25.7%), injection site swelling (22.1%), injection site induration (17.6%), and chills (11.2%).

In children from 6 to 35 months of age, the most frequently reported adverse reactions after vaccination were: injection site pain/tenderness (29.4%), fever (20.4%) and injection site erythema (17.2%).

- In the subpopulation of children from 6 to 23 months of age, the most frequently reported adverse reactions after vaccination were: irritability (34.9%), abnormal crying (31.9%), lost appetite (28.9%), somnolence (19.2%) and vomiting (17.0%).
- In the subpopulation of children from 24 to 35 months of age, the most frequently reported adverse reactions after vaccination were: malaise (26.8%), myalgia (14.5%) and headache (11.9%).

Table 2: Adverse reactions reported in children and adolescents from 6 months to 17 years of age

System Organ Class (SOC)/ Adverse Reactions	**Frequency**			
	Children 6-35 months of age		**Children 3-8 years of age**	**Children and adolescents 9-17 years of age**
	6-23 months of age	**24-35 months of age**		
Blood and Lymphatic System Disorders				
- Lymphadenopathy	Not known*		Uncommon	Not known*

System Organ Class (SOC)/ Adverse Reactions	Frequency			
	Children 6-35 months of age		Children 3-8 years of age	Children and adolescents 9-17 years of age
	6-23 months of age	24-35 months of age		
- Thrombocytopenia	Not known*		Uncommon	Not known*
Immune System Disorders				
- Allergic reactions such as:				
• Oropharyngeal pain	-		Uncommon	-
• Hypersensitivity	Uncommon		-	-
• Rash	-		Uncommon	Uncommon
• Urticaria	Not known*		Uncommon	Uncommon
• Pruritus	Not known*		Uncommon	Not known*
• Generalised pruritus, papular rash	Rare		-	-
• Erythematous rash, dyspnoea, anaphylactic reaction, angioedema, shock	Not known*		Not known*	Not known*
Metabolism and Nutrition Disorders				
- Decreased appetite	Very common	Rare	-	-
Psychiatric Disorders				
- Abnormal crying	Very common	-	-	-
- Irritability	Very common	Rare	-	-
- Agitation	-		Uncommon	-
- Moaning	-		Uncommon	-
Nervous System Disorders				
- Headache	-	Very common	Very common	Very common
- Somnolence	Very common	-	-	-
- Dizziness	-		Uncommon	Uncommon
- Neuralgia, neuritis and Guillain Barré Syndrome	-		Not known*	Not known*
- Paraesthesia, convulsions, encephalomyelitis	Not known*		Not known*	Not known*
Vascular Disorders				

System Organ Class (SOC)/ Adverse Reactions	Frequency			
	Children 6-35 months of age		Children 3-8 years of age	Children and adolescents 9-17 years of age
	6-23 months of age	24-35 months of age		
- Vasculitis such as Henoch-Schonlein purpura, with transient renal involvement in certain cases	Not known*		Not known*	Not known*
Gastrointestinal Disorders				
- Diarrhoea	Common		Uncommon	Uncommon
- Abdominal pain	-		Uncommon	-
- Vomiting	Very common	Uncommon	Uncommon	-
Musculoskeletal and Connective Tissue Disorders				
- Myalgia	Rare	Very common	Very common	Very common
- Arthralgia	-		Uncommon	-
General Disorders and Administration Site Conditions				
Injection site reactions				
- Injection site pain/tenderness, injection site erythema	Very common		Very common	Very common
- Injection site swelling	Common		Very common	Very common
- Injection site induration	Common		Very common	Common
- Injection site ecchymosis	Common		Common	Common
- Injection site pruritus	Rare		Uncommon	Uncommon
- Injection site warmth	-		Uncommon	Uncommon
- Injection site discomfort	-		-	Uncommon
- Injection site rash	Rare		-	-
Systemic reactions				
- Malaise	Rare	Very common	Very common	Very common
- Chills	-	Common	Very common	Very common
- Fever	Very common		Common	Common
- Fatigue	-		Uncommon	Uncommon

System Organ Class (SOC)/ Adverse Reactions	Frequency			
	Children 6-35 months of age		Children 3-8 years of age	Children and adolescents 9-17 years of age
	6-23 months of age	24-35 months of age		
- Asthenia	-		-	Uncommon
- Crying	-		Uncommon	-
- Influenza-like illness	Rare			-

(*) Adverse reactions reported post-marketing after use of Vaxigrip or Vaxigrip Tetra

Other special populations

- Although only a limited number of subjects with co-morbidities were enrolled, studies conducted in patients with co-morbidities such as renal transplant or asthmatic patients, showed no major differences in terms of safety profile of Vaxigrip and Vaxigrip Tetra in these populations.

Pregnant women

In clinical studies conducted in pregnant women in South Africa and Mali with Vaxigrip (see Sections 4.6 and 5.1), frequencies of local and systemic solicited reactions reported within 7 days following administration of the vaccine, were consistent with those reported for the adult population during clinical studies. In the study conducted in South Africa, local reactions were more frequent in the Vaxigrip group than in the placebo group in both HIV-negative and HIV-positive cohorts. There were no other significant differences in solicited reactions between Vaxigrip and placebo groups in both cohorts.

In one clinical study conducted in pregnant women in Finland with Vaxigrip and Vaxigrip Tetra (see Sections 4.6 and 5.1), frequencies of reported local and systemic solicited reactions were consistent with those reported for the non-pregnant adult population during clinical studies conducted with Vaxigrip or Vaxigrip Tetra even though higher for some adverse reactions (injection site pain, injection site erythema, malaise, chills, headache, myalgia).

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system.

4.9 Overdose

Cases of administration of more than the recommended dose (overdose) have been reported with Vaxigrip. When adverse reactions were reported, the information was consistent with the known safety profile of Vaxigrip described in Section 4.8.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic Properties

Pharmacotherapeutic group: influenza vaccine

ATC code: J07BB02

Mechanism of action

Vaxigrip provides active immunisation against three influenza virus strains (two A subtypes and one B type) contained in the vaccine.

Vaxigrip induces humoral antibodies against the haemagglutinins within 2 to 3 weeks. These antibodies neutralise influenza viruses.

In infants less than 6 months of age born to women vaccinated with Vaxigrip during pregnancy, protection is due to transplacental transfer of these neutralizing antibodies.

There is no correlation between specific levels of antibody titres after vaccination with inactivated influenza virus vaccines , measured by haemagglutination inhibition (HAI), with protection against influenza, but HAI antibody titres have been used as a measure of vaccine activity. In some human studies, HAI antibody titres ≥1/40 have been associated with protection against influenza in up to 50% of subjects.

Since influenza viruses constantly evolve, the virus strains selected for the vaccine are reviewed annually by the WHO.

Annual influenza vaccination is recommended given the duration of immunity provided by the vaccine and because circulating strains of influenza virus change from year to year.

Efficacy

Efficacy data of Vaxigrip are available in pregnant women and in infants less than 6 months of age born to vaccinated pregnant women (passive protection).
No efficacy studies have been performed with Vaxigrip in children and adolescents from 9 to 17 years of age, in adults and in the elderly.
In children from 6 to 35 months of age and from 3 to 8 years of age (active immunisation), Vaxigrip efficacy is based on extrapolation of Vaxigrip Tetra efficacy.

- *Infants less than 6 months of age born to women vaccinated during their pregnancies (passive protection):*

Infants less than 6 months of age are at high risk of influenza, resulting in high rates of hospitalisation. However, influenza vaccines are not indicated for active immunisation in this age group.

Efficacy in infants born of women who received a single 0.5 mL dose of Vaxigrip during the second or third trimester of pregnancy has been demonstrated in clinical trials.
The efficacy of Vaxigrip in infants born to women vaccinated during the first trimester of pregnancy was not studied in these trials. If influenza vaccination is considered necessary during the first trimester of pregnancy, it should not be postponed (see Section 4.6).

In randomized, controlled phase IV clinical studies conducted in Mali, Nepal and South Africa, approximately 5,000 pregnant women received Vaxigrip and approximately 5,000 pregnant women received placebo or control vaccine (quadrivalent meningococcal conjugate vaccine) during the second or third trimester of pregnancy. Vaccine efficacy in the prevention of laboratory confirmed influenza in pregnant women was evaluated as a secondary endpoint in all three studies.

The studies conducted in Mali and South Africa demonstrated the efficacy of Vaxigrip in the prevention of influenza in pregnant women following vaccination during these trimesters of pregnancy (see Table 3). In the study conducted in Nepal, the efficacy of Vaxigrip in the prevention of influenza in pregnant women following vaccination during these trimesters of pregnancy was not demonstrated.

Table 3: Influenza Attack Rates and Vaxigrip Efficacy against Laboratory-confirmed influenza in pregnant women

	Influenza Attack Rate (Whether type A or B) % (n/N)		Vaxigrip Efficacy % (95% CI)
	Vaxigrip	**Control***	
Mali	0.5 (11/2,108)	1.9 (40/2,085)	70.3 (42.2; 85.8)
	Vaxigrip	**Placebo**	
South Africa	1.8 (19/1,062)	3.6 (38/1,054)	50.4 (14.5; 71.2)

* Meningococcal vaccine
N: Number of pregnant women included in analysis
n: number of subjects with laboratory confirmed influenza
CI: Confidence Interval

In the same randomized, controlled phase IV clinical studies conducted in Mali, Nepal and South Africa, 4530 of 4898 (92%) infants born to pregnant women who received Vaxigrip and 4532 of 4868 (93%) infants born to pregnant women who received a placebo or control vaccine (quadrivalent meningococcal conjugate vaccine) (see Table 4) during the second or third trimester of pregnancy, were followed up until approximately 6 months of age.

The studies confirmed the efficacy of Vaxigrip in the prevention of influenza in infants born to women vaccinated during these trimesters of pregnancy, from birth until approximately 6 months of age. Women in their first trimester of pregnancy were not included in these studies; Vaxigrip efficacy in infants born to mothers vaccinated during the first trimester could therefore not be evaluated.

Table 4: Influenza Attack Rates and Vaxigrip Efficacy against Laboratory-confirmed influenza in infants born to women vaccinated during their pregnancies

	Influenza Attack Rate (Whether type A or B) % (n/N)		Vaxigrip Efficacy % (95% CI)
	Vaxigrip	**Control***	
Mali	2.4 (45/1,866)	3.8 (71/1,869)	37.3 (7.6; 57.8)
	Vaxigrip	**Placebo**	
Nepal	4.1 (74/1,820)	5.8 (105/1,826)	30.0 (5; 48)
South Africa	1.9 (19/1,026)	3.6 (37/1,023)	48.8 (11.6; 70.4)

* Meningococcal vaccine
N: Number of infants included in the analysis
n: number of subjects with laboratory-confirmed influenza
CI: Confidence Interval

The efficacy data indicate a decrease over time after birth in the protection of infants born to mothers vaccinated during their pregnancies.

In the trial conducted in South Africa, vaccine efficacy was higher in infants 8 weeks of age or younger (85.8% [95% CI: 38.3; 98.4]) and decreased over time; vaccine efficacy was 25.5% (95% CI: -67.9; 67.8) in infants from 8 to 16 weeks of age and 30.4% (95% CI: -154.9; 82.6) in infants from 16 to 24 weeks of age.

In the trial conducted in Mali, there was also a trend for higher efficacy of Vaxigrip in infants during the first four months after birth, with lower efficacy within the 5th month of surveillance and a marked fall within the 6th month when protection is no longer evident.

The prevention of influenza can only be expected if the infant(s) are exposed to strains included in the vaccine administered to the mother.

- *Children from 6 to 35 months of age (active immunisation):*

A randomized placebo controlled study was conducted in 4 regions (Africa, Asia, Latin America and Europe) over 4 influenza seasons, in more than 5,400 children from 6 to 35 months of age who received two doses (0.5 mL) of Vaxigrip Tetra (N=2,722), or placebo (N=2,717) 28 days apart to assess Vaxigrip Tetra efficacy in the prevention of laboratory-confirmed influenza caused by any strain A and/or B and caused by strains similar to those of the vaccine (as determined by sequencing).

Laboratory-confirmed influenza was defined as influenza like-illness (ILI) [occurrence of fever ≥ 38°C (lasting at least 24 hours) concurrently with at least one of the following symptoms: cough, nasal congestion, rhinorrhoea, pharyngitis, otitis, vomiting, or diarrhoea], laboratory-confirmed by reverse transcriptase polymerase chain reaction (RT-PCR) and/or viral culture.

Table 5: Influenza Attack Rates and Vaxigrip Tetra Efficacy against laboratory-confirmed influenza in children from 6 to 35 months of age

	Vaxigrip Tetra (N=2,584)		**Placebo (N=2,591)**		**Efficacy**
	N	**Influenza Attack Rate (%)**	**n**	**Influenza Attack Rate (%)**	**% (2-sided 95% CI)**
Laboratory-confirmed influenza caused by:					
- Any type A or B influenza	122	4.72	255	9.84	52.03 (40.24; 61.66)
- Viral strains similar to those contained in the vaccine	26	1.01	85	3.28	69.33 (51.93; 81.03)

N: Number of children analysed (full set)
n: number of children fulfilling the listed criteria
CI: Confidence Interval

In addition, a predefined complementary analysis showed Vaxigrip Tetra prevented 56.6% (95% CI: 37.0; 70.5) of severe laboratory-confirmed influenza due to any strain, and 71.7% (95% CI: 43.7; 86.9) of severe laboratory-confirmed influenza due to strains similar to those of the vaccine. Furthermore, subjects receiving Vaxigrip Tetra were 59.2% (95% CI: 44.4; 70.4) less likely to experience influenza requiring a medical consultation than subjects receiving placebo.

Severe laboratory-confirmed influenza was defined as influenza-like illness laboratory-confirmed by RT-PCR and/or viral culture with at least one of the following:

- fever > 39.5°C for subjects aged < 24 months or ≥ 39.0°C for subjects aged ≥ 24 months,
- and/or at least one significant flu-like symptom which prevents daily activity (cough, nasal congestion, rhinorrhoea, pharyngitis, otitis, vomiting, diarrhoea),
- and/or one of the following events: acute otitis media, acute lower respiratory infection (pneumonia, bronchiolitis, bronchitis, croup), inpatient hospitalisation.

- *Children from 3 to 8 years of age (active immunisation):*

Based on immune responses of Vaxigrip observed in children 3 to 8 years of age, the efficacy of Vaxigrip in this population is expected to be at least similar to the efficacy observed in children from 6 to 35 months (see "*Children from 6 to 35 months of age(active immunisation)*" above and "Immunogenicity of Vaxigrip" below).

Immunogenicity

Clinical studies performed in adults from 18 to 60 years of age, in the elderly over 60 years of age, and in children from 3 to 8 years of age and from 6 to 35 months of age, evaluated the immune response to Vaxigrip (TIV) and Vaxigrip Tetra (QIV) as regards the geometric mean titres (GMTs) in IHA antibodies at Day 21 (for adults) and at Day 28 (for children), the HAI seroconversion rate (4-fold rise in reciprocal titre or change from an undetectable titre [< 10] to a reciprocal titre ≥ 40), and the geometric mean of individual titre ratios (GMTRs) of HAI (post-/pre-vaccination titres).

One clinical study performed in adults from 18 to 60 years of age and in children from 9 to 17 years of age described the immune response of Vaxigrip Tetra and Vaxigrip as regards the GMT in HAI antibodies on Day 21. Another clinical study performed in children from 9 to 17 years of age described the immune response of Vaxigrip Tetra.

One clinical study performed in pregnant women described the immune response of Vaxigrip and Vaxigrip Tetra as regards the GMT in for HAI antibodies at Day 21, the HAI seroconversion rate, and the HAI GMTR after one dose administered during the second or third trimester of pregnancy. In this study, transplacental transfer was evaluated using GMTs in HAI antibodies of maternal blood, of cord blood and the ratio of cord blood/maternal blood, at delivery.

Overall, Vaxigrip induced an immune response to the 3 influenza strains contained in the vaccine.

In children from 3 years of age and in adults including pregnant women and the elderly, Vaxigrip was as immunogenic as Vaxigrip Tetra for the strains in common.

Antibody persistence was assessed in adults, the elderly and children aged 6 months to 35 months. The duration of the immunity induced by vaccination is at least 12 months.

- *Adults and the elderly*

In one clinical study, the immune response was described in adults from 18 to 60 years of age and the elderly over 60 years of age who received one 0.5-mL dose of Vaxigrip or Vaxigrip Tetra.

The immunogenicity results by HAI method in adults from 18 to 60 years of age and elderly over 60 years of age are presented in Table 6.

Table 6: Immunogenicity results in adults from 18 to 60 years of age and in the elderly over 60 years of age, 21 days after vaccination with Vaxigrip or Vaxigrip Tetra

	Adults from 18 to 60 years of age			Elderly over 60 years of age		
Antigen Strain	Alternative TIV [(a)] (B Victoria) N=140	Licensed TIV [(b)] (B Yamagata) N=138	QIV N=832	Alternative TIV [(a)] (B Victoria) N=138	Licensed TIV [(b)] (B Yamagata) N=137	QIV N=831
	GMT (95% CI)					
A (H1N1) [(c)(d)]	685 (587; 800)		608 (563;657)		268 (228; 314)	219 (199; 241)
A (H3N2) [(c)]	629 (543; 728)		498 (459; 541)		410 (352; 476)	359 (329; 391)
B (Victoria)	735 (615; 879)	-	708 (661; 760)	301 (244; 372)	-	287 (265; 311)
B (Yamagata)	-	1735 (1490; 2019)	1715 (1607; 1830)	-	697 (593; 820)	655 (611; 701)
	% SC (95% CI) [(e)]					
A (H1N1) [(c)(d)]	65.1 (59.2; 70.7)		64.1 (60.7; 67.4)		50.2 (44.1; 56.2)	45.6 (42.1; 49.0)
A (H3N2) [(c)]	73.4 (67.8; 78.5)		66.2 (62.9; 69.4)		48.5 (42.5; 54.6)	47.5 (44.1; 51.0)
B (Victoria)	70.0 (61.7; 77.4)	-	70.9 (67.7; 74.0)	43.5 (35.1; 52.2)	-	45.2 (41.8; 48.7)
B (Yamagata)	-	60.9 (52.2; 69.1)	63.7 (60.3;67.0)	-	38.7 (30.5; 47.4)	42.7 (39.3; 46.2)
	GMTR (95% CI) [(f)]					
A (H1N1) [(c)(d)]	10.3 (8.35; 12.7)		9.77 (8.69; 11.0)		6.03 (4.93; 7.37)	4.94 (4.46; 5.47)
A (H3N2) [(c)]	14.9 (12.1; 18.4)		10.3 (9.15; 11.5)		5.79 (4.74; 7.06)	5.60 (5.02; 6.24)

Antigen Strain	Adults from 18 to 60 years of age			Elderly over 60 years of age		
	Alternative TIV (a) (B Victoria) N=140	**Licensed TIV (b) (B Yamagata) N=138**	**QIV N=832**	**Alternative TIV (a) (B Victoria) N=138**	**Licensed TIV (b) (B Yamagata) N=137**	**QIV N=831**
B (Victoria)	11.4 (8.66; 15.0)	-	11.6 (10.4; 12.9)	4.60 (3.50; 6.05)	-	4.61 (4.18; 5.09)
B (Yamagata)	-	6.08 (4.79; 7.72)	7.35 (6.66;8.12)	-	4.11 (3.19; 5.30)	4.11 (3.73; 4.52)

N: number of subjects with available data for the considered endpoint
GMT: Geometric mean titre; CI: Confidence Interval
(a) Alternative TIV containing A/California/7/2009 (H1N1), A/Texas/50/2012 (H3N2), and B/Brisbane/60/2008 (Victoria lineage)
(b) 2014-2015 licensed TIV containing A/California/7/2009 (H1N1), A/Texas/50/2012 (H3N2), and B/Massachusetts/2/2012 (Yamagata lineage)
(c) The pooled TIV group includes participants vaccinated with either an alternative TIV or a licensed TIV, N= 278 for adults aged 18 to 60 years and N=275 for the elderly aged over 60 years
(d) N=833 for the QIV group in adults aged 18 to 60 years; N=832 for the QIV group in the elderly aged over 60 years
(e) SC: Seroconversion or significant increase: for subjects with a pre-vaccination titre <10 (1/dil), proportion of subjects with a post-vaccination titre ≥40 (1/dil) and for subjects with a pre-vaccination titre ≥10 (1/dil), proportion of subjects with a ≥four-fold increase in the titre pre- to post-vaccination
(f) GMTR: Geometric mean of individual titre ratios (post-/pre-vaccination titres)

- *Pregnant women and transplacental transfer*

In one clinical study, a total of 116 pregnant women received Vaxigrip and 230 pregnant women received Vaxigrip Tetra during the second or third trimester of pregnancy (from 20 to 32 weeks of pregnancy).

Immunogenicity results by HAI method, in pregnant women 21 days after vaccination with Vaxigrip or Vaxigrip Tetra are presented in Table 7.

Table 7: Immunogenicity results by HAI method in pregnant women, 21 days after vaccination with Vaxigrip or Vaxigrip Tetra

Antigen Strain	**TIV (B Victoria) N=109**	**QIV N=216**
	GMT (95% CI)	
A (H1N1)*	638 (529; 769)	525 (466; 592)
A (H3N2)*	369 (283; 483)	341 (286; 407)
B1 (Victoria)*	697 (569; 855)	568 (496; 651)
B2 (Yamagata)*	-	993 (870; 1134)
	≥4-fold-rise n (%) (a)	
A (H1N1)*	41.3 (31.9; 51.1)	38.0 (31.5; 44.8)

Antigen Strain	TIV (B Victoria) N=109	QIV N=216
A (H3N2)*	62.4 (52.6; 71.5)	59.3 (52.4; 65.9)
B1 (Victoria)*	60.6 (50.7; 69.8)	61.1 (54.3; 67.7)
B2 (Yamagata)*	-	59.7 (52.9; 66.3)
	GMTR (95% CI) (b)	
A (H1N1)*	5.26 (3.66; 7.55)	3.81 (3.11; 4.66)
A (H3N2)*	9.23 (6.56; 13.0)	8.63 (6.85; 10.9)
B1 (Victoria)*	9.62 (6.89; 13.4)	8.48 (6.81; 10.6)
B2 (Yamagata)*	-	6.26 (5.12; 7.65)

N: number of subjects with available data for the considered endpoint
GMT: Geometric mean titre; CI: Confidence Interval
*A/H1N1: A/Michigan/45/2015 (H1N1) pdm09-like strain; A/H3N2: A/Hong Kong/4801/2014 (H3N2)-like strain;
B1: B/Brisbane/60/2008-like strain (B/Victoria lineage)*: this strain was included in the TIV composition;*
B2: B/Phuket/3073/2013-like strain (B/Yamagata lineage)*: this strain was not included in the TIV composition.*
(a) SC: Seroconversion or significant increase: for subjects with a pre-vaccination titre <10 (1/dil), proportion of subjects with a post-vaccination titre ≥40 (1/dil) and for subjects with a pre-vaccination titre ≥10 (1/dil), proportion of subjects with a ≥four-fold pre- to post-vaccination increase in the titre
(b) GMTR: Geometric mean of individual titre ratios (post-/pre-vaccination titres)

A descriptive analysis of immunogenicity by HAI method, at delivery, in a sample of mother's blood (BL03M), and in a sample of cord blood (BL03B) and of transplacental transfer (BL03B/ BL03M) are presented in Table 8.

Table 8: Descriptive analysis of immunogenicity by the HAI method with Vaxigrip or Vaxigrip Tetra, at delivery

Antigen Strain	TIV (B Victoria) N=89	QIV N=178
	BL03M (Maternal blood) GMT (95% CI)	
A (H1N1)*	411 (332; 507)	304 (265; 349)
A (H3N2)*	186 (137; 250)	178 (146; 218)
B1 (Victoria)*	371 (299; 461)	290 (247; 341)
B2 (Yamagata)*	-	547 (463; 646)
	BL03B (Cord blood) GMT (95% CI)	
A (H1N1)*	751 (605; 932)	576 (492; 675)
A (H3N2)*	324 (232; 452)	305 (246; 379)
B1 (Victoria)*	608 (479; 772)	444 (372; 530)
B2 (Yamagata)*	-	921 (772; 1099)
	Transplacental transfer: BL03B/BL03M GMT (95% CI)**	

Antigen Strain	TIV (B Victoria) N=89	QIV N=178
A (H1N1)*	1.83 (1.64; 2.04)	1.89 (1.72; 2.08)
A (H3N2)*	1.75 (1.55; 1.97)	1.71 (1.56; 1.87)
B1 (Victoria)*	1.64 (1.46; 1.85)	1.53 (1.37; 1.71)
B2 (Yamagata)*	-	1.69 (1.54; 1.85)

N: number of subjects with available data for the considered endpoint: women who received QIV or TIV, who gave birth at least 2 weeks after the injection, and with cord blood and mother's blood available at the time of delivery.
GMT: Geometric mean titre; CI: Confidence interval
* A/H1N1: A/Michigan/45/2015 (H1N1) pdm09-like strain;
A/H3N2: A/Hong Kong/4801/2014 (H3N2)-like strain;
B1: B/Brisbane/60/2008-like strain (B/Victoria lineage): this strain was included in the TIV composition;
B2: B/Phuket/3073/2013-like strain (B/Yamagata lineage): this strain was not included in the TIV composition
** If a mother has X babies, her titre values are counted X times

At delivery, the higher level of antibodies in the cord sample compared to the maternal sample is consistent with transplacental antibody transfer from mother to the foetus following vaccination of women with Vaxigrip or Vaxigrip Tetra during the second or third trimester of pregnancy.

These data are consistent with the passive protection demonstrated in infants from birth to approximately 6 months of age born of women vaccinated during the second or third trimester of pregnancy with Vaxigrip in studies conducted in Mali, Nepal, and South Africa (see subsection "Efficacy")

- *Paediatric population*

- Children from 9 to 17 years of age

In a total of 55 children from 9 to 17 years of age who received one dose of Vaxigrip and 429 who received one dose of Vaxigrip Tetra, the immune response against the strains contained in the vaccine was similar to the immune response induced in adults 18 to 60 years of age.

- Children from 3 years to 8 years of age

In one clinical study, the immune response was described in children from 3 to 8 years of age who received either one or two 0.5 mL doses of Vaxigrip or Vaxigrip Tetra, depending on their previous influenza vaccination history.
Children who received a one- or two-dose schedule of Vaxigrip or Vaxigrip Tetra presented a similar immune response following the last dose of the respective schedule.

The immunogenicity results by HAI method 28 days after receipt of the last injection are presented in Table 9.

- Children from 6 months to 35 months of age

In one clinical trial, the immune response was described in children from 6 to 35 months of age who received two 0.5-mL doses of Vaxigrip or Vaxigrip Tetra.

The immunogenicity results by HAI method, 28 days after receipt of the last injection are presented in Table 9.

Table 1:Immunogenicity results in children from 6 months to 35 months of age and from 3 to 8 years of age, 28 days after the last injection of Vaxigrip or Vaxigrip Tetra

	Children from 6 to 35 months of age			Children from 3 to 8 years of age		
Antigen Strain	**Alternative TIV [(a)] (B Victoria) N=172**	**Licensed TIV [(b) (c)] (B Yamagata) N=178**	**QIV N=341**	**Alternative TIV [(a)] (B Victoria) N=176**	**Licensed TIV [(b)] (B Yamagata) N=168**	**QIV N=863**
	GMT (95% CI)					
A (H1N1) [(d)]	637 (500; 812)	628 (504; 781)	641 (547; 752)	1141 (1006; 1295)		971 (896; 1052)
A (H3N2) [(d)]	1021 (824; 1266)	994 (807; 1224)	1071 (925; 1241)	1746 (1551; 1964)		1568 (1451; 1695)
B (Victoria) [(e)]	835 (691; 1008)	-	623 (550; 706)	1120 (921; 1361)	-	1050 (956; 1154)
B (Yamagata) [(f) (g)]	-	1009 (850; 1198)	1010 (885; 1153)	-	1211 (1003; 1462)	1173 (1078; 1276)
	% SC (95% CI) [(e)]					
A (H1N1) [(d)]	87.2 (81.3; 91.8)	90.4 (85.1; 94.3)	90.3 (86.7; 93.2)	65.7 (60.4; 70.7)		65.7 (62.4; 68.9)
A (H3N2) [(d)]	88.4 (82.6; 92.8)	87.6 (81.9; 92.1)	90.3 (86.7; 93.2)	67.7 (62.5; 72.6)		64.8 (61.5; 68.0)
B (Victoria) [(e)]	99.4 (96.8; 100.0)	-	98.8 (97.0; 99.7)	90.3 (85.0; 94.3)	-	84.8 (82.3; 87.2)
B (Yamagata) [(f) (g)]	-	99.4 (96.9; 100.0)	96.8 (94.3; 98.4)	-	89.9 (84.3; 94.0)	88.5 (86.2; 90.6)
	GMTR (95% CI) [(f)]					
A (H1N1) [(d)]	35.3 (27.4; 45.5)	40.6 (32.6; 50.5)	36.6 (30.8; 43.6)	7.65 (6.54; 8.95)		6.86 (6.24; 7.53)
A (H3N2) [(d)]	44.1 (33.1; 58.7)	37.1 (28.3; 48.6)	42.6 (35.1; 51.7)	7.61 (6.69; 9.05)		7.49 (6.72; 8.35)
B (Victoria) [(e)]	114 (94.4; 138)	-	100 (88.9; 114)	17.8 (14.5; 22.0)	-	17.1 (15.5; 18.8)
B (Yamagata) [(f) (g)]	-	111 (91.3; 135)	93.9 (79.5; 111)	-	30.4 (23.8; 38.4)	25.3 (22.8; 28.2)

N=number of subjects with available data for the considered endpoint
GMT: Geometric mean titre; CI: Confidence Interval
(a) Alternative TIV containing A/California/7/2009 (H1N1), A/Texas/50/2012 (H3N2), and B/Brisbane/60/2008 (Victoria lineage)
(b) 2014-2015 licensed TIV containing A/California/7/2009 (H1N1), A/Texas/50/2012 (H3N2), and B/Massachusetts/2/2012 (Yamagata lineage)
(c) Dose of 0.5 mL in children 6 to 35 months of age

(d) For children -3 to 8 years of age: The TIV group includes participants vaccinated with either alternative Vaxigrip or licensed TIV, N=344
(e) N=169 for Vaxigrip (B Victoria) group in children 3 to 8 years of age
(f) N=862 for QIV group in children 3 to 8 years of age
(g) For alternative TIV (B Victoria) group: N=171 for children 6 to 35 months of age; N=175 for children 3 to 8 years of age
(h) SC: Seroconversion or significant increase: for subjects with a pre-vaccination titre <10 (1/dil), proportion of subjects with a post-vaccination titre ≥40 (1/dil) and for subjects with a pre-vaccination titre ≥10 (1/dil), proportion of subjects with a ≥four-fold increase in the titre post-vaccination
(i) GMTR: Geometric mean of individual titre ratios (post-/pre-vaccination titres)

These immunogenicity data provide supportive information in addition to efficacy data available in children from 6 to 35 months of age (see Section "Efficacy").

5.2 Pharmacokinetic Properties

Not applicable.

5.3 Preclinical Safety Data

Data in animals generated with Vaxigrip Tetra (60 µg of total HA/dose) can be extrapolated to Vaxigrip (45 µg of total HA/dose): these data revealed no unexpected findings and no target organ toxicity.

6. PHARMACEUTICAL PARTICULARS

6.1 List of Excipients

Buffer solution:

- Sodium chloride
- Potassium chloride
- Disodium phosphate dihydrate
- Potassium dihydrogen phosphate
- Water for injections

6.2 Incompatibilities

In the absence of compatibility studies, this vaccine must not be mixed with other medicinal products.

6.3 Shelf Life

1 year.

6.4 Special Precautions for Storage

Store in a refrigerator (2°C – 8°C). Do not freeze. Keep the syringe in the outer carton in order to protect from light.

Vaxigrip remains stable for 72 hours at temperatures up to 25°C ±2°C. This is not a recommended storage condition; it is only intended to guide healthcare professionals in case of temporary temperature excursion.

6.5 Nature and Contents of Container

- 0.5 mL of suspension in pre-filled syringe (type I glass) with attached needle fitted with a plunger stopper (bromobutyl elastomer) – box of 1 or 10.
- 0.5 mL of suspension in pre-filled syringe (type I glass) fitted with a plunger stopper (bromobutyl elastomer) and a tip cap:
    - Box of 1 or 10 pre-filled syringe(s) without needle(s)
    - Box of 1 or 10 pre-filled syringe(s) with separate needle(s)
    - Box of 1 or 10 pre-filled syringe(s) with separate needle(s) with a safety shield.

Not all pack sizes may be marketed.

6.6 Special Precautions for Disposal and Other Handling

The vaccine should be allowed to reach room temperature before use.

Shake before use.

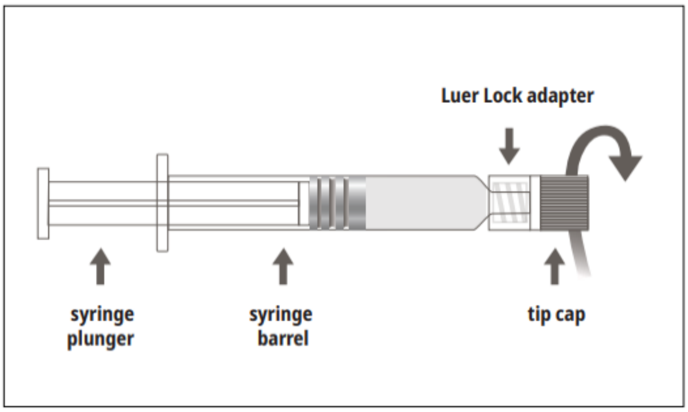
Preparation for administration:

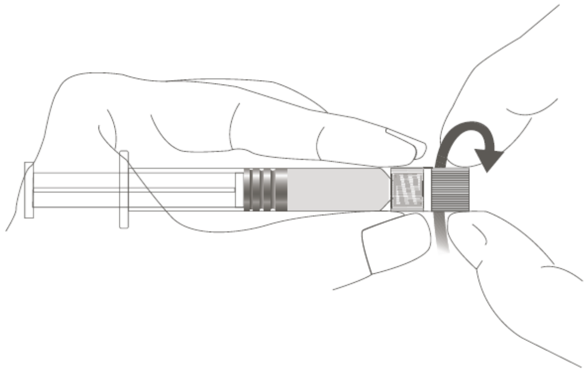
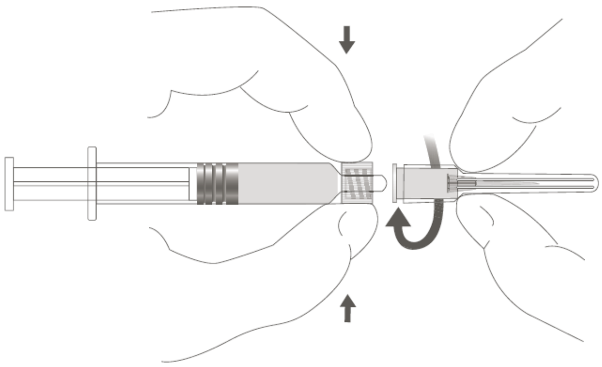
The syringe containing the suspension for injection must be visually inspected before administration. In the event of foreign particles, leakage, premature activation of the plunger or a faulty tip, discard the pre-filled syringe.

The syringe is intended for single use only and must not be reused.

Instructions for use for the luer lock pre-filled syringe.

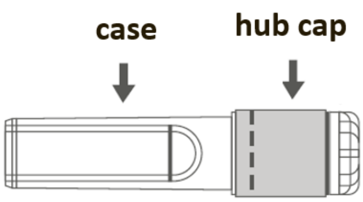
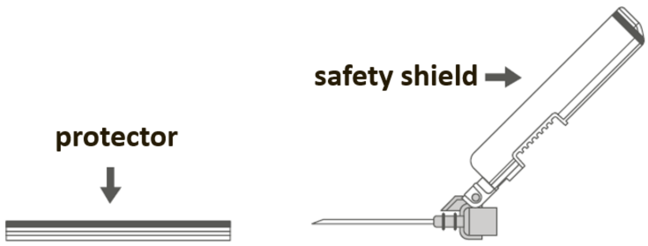
Picture A: Luer Lock syringe with Rigid Tip Cap



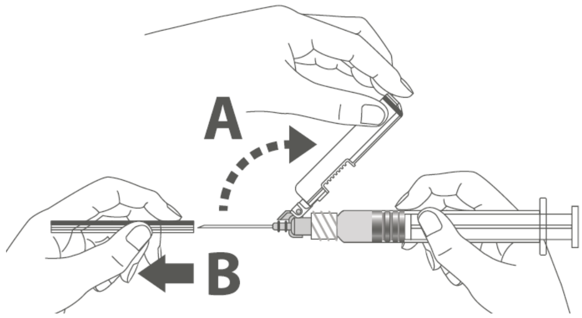
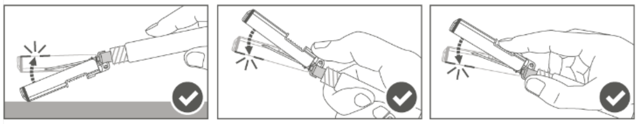
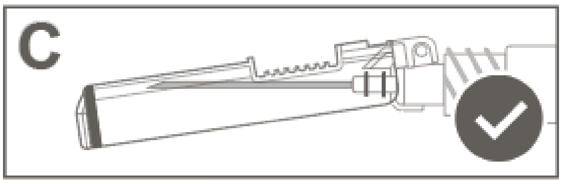
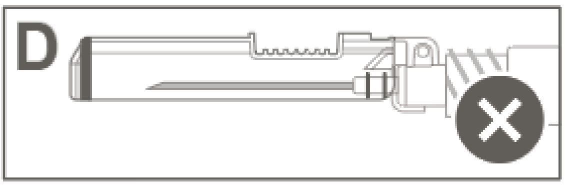
Step 1: Holding the Luer Lock adapter in one hand (avoid holding the syringe plunger or barrel), unscrew the tip cap by twisting it.	
Step 2: To attach the needle to the syringe, gently twist the needle into the Luer Lock adapter of the syringe until slight resistance is felt.	

<Instructions for use of Safety Needle with Luer Lock pre-filled syringe:

Follow Steps 1 and 2 above to prepare the Luer Lock syringe and needle for attachment.

Picture B: Safety Needle (inside case)	**Picture C: Safety Needle Components (prepared for use)**
	

Step 3: Pull the safety needle's case straight off. The needle is covered by the safety shield and protector.

Step 4: **A:** Move the safety shield away from the needle and toward the syringe barrel to the angle shown. **B:** Pull the protector straight off.	
Step 5: After injection is complete, lock (activate) the safety shield using one of the three (3) **one-handed** techniques illustrated: surface, thumb or finger activation. Note: Activation is verified by an audible and/or tactile "click."	
Step 6: Visually inspect the safety shield activation. The safety shield should be **fully locked (activated)** as shown in Figure C. Figure D shows the safety shield is **NOT fully locked (not activated).**	 
Caution: Do not attempt to unlock (deactivate) the safety device by forcing the needle out of the safety shield.>	

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. PRODUCT REGISTRATION HOLDER

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8. DATE OF REVISION OF THE TEXT

Mar 2026

(CCDS V15 based on FR SmPC 27 May 2025, SH2026)